

Enabling Routing and Spectrum Assignment with Parallel Links in the TeraFlow SDN Ecosystem

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Abstract—Flexible-grid optical networks improve spectrum efficiency but introduce additional challenges for Routing and Spectrum Allocation (RSA). These constraints become even more critical in the presence of parallel links between network nodes, where multiple links interconnect the same pair of nodes, causing a combinatorial growth of feasible routing options and making conventional path computation inefficient. This is further complicated by spectrum continuity and contiguity constraints requiring fine-grained slot management.

This paper presents the *ParallelOpticalController*, a microservice integrated into *TeraFlow* SDN ecosystem to address both routing scalability and spectrum allocation complexity. The solution models the network as a multi-directed graph and employs a bitwise algorithm to rapidly identify contiguous spectrum slots across end-to-end paths.

With respect to solutions adopted in other SDN controllers, the proposed approach reduces computational overhead while improving resource utilization, enabling scalable control of topologies with parallel links. The work investigates key design parameters, such as the number of candidate paths and spectrum assignment policies, to identify the trade-offs that provide the best balance between scalability and resource utilization.

Index Terms—Software-defined optical networks (SDON); Elastic Optical Network (EON); Parallel links.

I. INTRODUCTION

The paradigm shift in global data consumption, driven by bandwidth-intensive applications such as real-time ultra-high-definition streaming, large-scale High-Performance Computing, and the pervasive deployment of AI agents, has placed unprecedented pressure on network infrastructures [1]. To accommodate this growth, optical transport networks transitioned from traditional fixed-grid Wavelength Division Multiplexing (WDM) toward more versatile Elastic Optical Networks (EONs) based on a flexible grid: under the ITU-T G.694.1 standard [2], the optical spectrum is discretized into small Frequency Slot Units (FSUs). The nominal central frequency is defined by the formula: $f_c = 193.1THz + (n \times 0.00625)THz$, where n is an integer number. The width of the allocated slot (Δf) is determined by: $\Delta f = 12.5GHz \times m$, where m is a positive integer representing the number of contiguous slots. This granularity enables diverse bit rates to coexist within the same fiber, maximizing the total throughput. While EONs offer significant flexibility, they introduce the Routing and Spectrum Assignment (RSA) problem, a complex computational challenge in optical network management.

The complexity of establishing lightpaths in a topology grows with respect to the number of nodes and edges and the problem is formally established as NP-hard [3]. Addressing this architectural challenge, Sgambelluri et al. [4] designed and implemented a cloud-native SDN controller within the TeraFlowSDN framework to explicitly support dynamic space division multiplexing (SDM) via parallel fibers and wideband optical-band switching. Their expanded architecture features a dedicated Optical component and Southbound Interface (SBI) drivers that communicate with OpenConfig-augmented multi-granular nodes using NETCONF and YANG models. To process connectivity requests across this complex domain, the controller utilizes a multi-dimensional FLEX-RSA engine that prioritizes band selection (C-band, L-band, then S-band) while performing joint allocation across routing, spatial fiber IDs, and contiguous spectrum slots. Complementing these SDN-based architectural extensions, Alam et al. in [5] proposed the Minimum Hops with Least Slots (MHLS) algorithm to optimize spectral occupancy by calculating demand-ordering parameters based on Frequency Slot Units (FSUs) and hop count. However, the scalability of these RSA solutions is challenged by the structural complexities of parallel link architectures. While parallel fibers connecting distinct ROADMs line degrees facilitate spatial frequency reuse, fibers terminating at a shared transponder add/drop WSS strictly inhibit it.

Traditional SDON controllers treat parallel links as discrete entities, executing recursive RSA sequentially across individual paths until a valid allocation is identified. In high-density environments, this exhaustive search triggers exponential time complexity, creating computational bottlenecks that severely limit control plane scalability and resource efficiency. To address these bottlenecks, this paper introduces an optimized *ParallelOpticalController* (POC) integrated into the TeraFlowSDN (TFS) ecosystem implementing shortest path algorithm and parallel link aware spectrum assignment.

II. BACKGROUND AND PROBLEM STATEMENT

The RSA problem in flex-grid (EONs) is an evolution of the Routing and Wavelength Assignment (RWA) problem found in traditional fixed-grid WDM systems [6]. The fundamental difference lies in the EON's ability to adaptively partition the optical spectrum into variable-sized slots to match specific bitrate demands. Common approaches to RSA involve utilizing

the K-Shortest Path algorithm for route discovery and the first-fit (FF) heuristic for spectrum assignment (SA). RSA problems are categorized into two scenarios, static where traffic demands are known in advance, and dynamic, where connection requests arrive stochastically [7]. The RSA algorithm must find a path and then allocate a specific number of frequency slots to satisfy the bandwidth requirement. This allocation is strictly governed by the spectrum continuity and contiguity constraints, which mandate that the assigned frequency slots remain identical across all traversed links and strictly adjacent to one another within the spectrum, respectively [8]. If a network controller fails to identify a block of frequency slots that satisfies both conditions, the lightpath request is blocked, regardless of the total aggregate free capacity. These fundamental constraints become significantly more complex with parallel links.

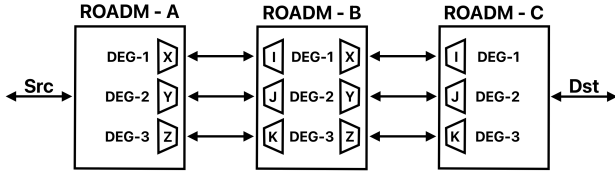


Fig. 1. Linear topology with 3 nodes and 3 parallel links between each pair.

As illustrated in Fig. 1, a lightpath request from *Src* to *Dst* traverses nodes A, B, and C, which are interconnected by three physically independent parallel fibers. Effective RSA must jointly determine the spatial path selection and the allocation of contiguous spectral slots to avoid blocking. Thus, for topologies with a uniform number of parallel links between adjacent nodes, the spatial search space scales as P^H for a path of length H hops with P links per hop. In non-uniform scenarios, the search space generalizes to $\prod_{k=1}^H P_k$, where P_k denotes the number of available links at hop k .

III. THE PARALLEL OPTICAL CONTROLLER

TFS is a state-of-the-art, open-source SDN controller developed under the European Telecommunications Standards Institute (ETSI) to radically advance the orchestration of beyond 5G and 6G networks [9]–[11]. Designed as a secure, cloud-native platform, TFS employs a microservices-based architecture orchestrated within Kubernetes environments, enabling individual control components to operate autonomously and scale independently. As illustrated in Fig. 2, the architecture is explicitly engineered to seamlessly integrate optical network functionality management with modern Network Function Virtualization and Multi-access Edge Computing frameworks, alongside robust load-balancing and security mechanisms.

A. Design

POC is deliberately designed as a decoupled, portable solution via MicroK8s [12]. As illustrated in Fig. 3, the initiation of a lightpath request via the TFS service component generates a service request object exposed to the POC. The POC constructs the network topology by querying relevant

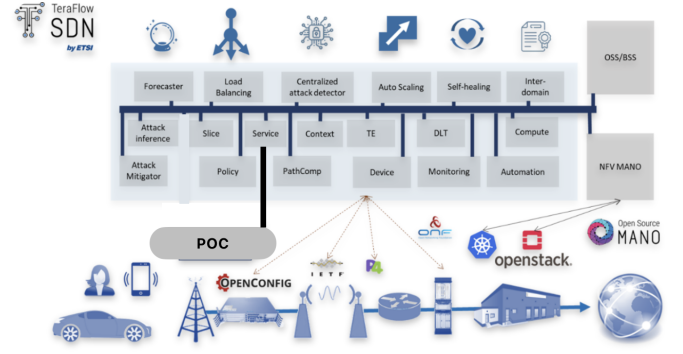


Fig. 2. TFS cloud-native architecture with core network functions and POC.

data from the TFS context and device services [4]. This topology is transmitted to the RSA Engine, which utilizes specialized functions in RSA Tools for graph generation and path computation. Upon identifying a suitable route, RSA Tools executes spectrum assignment encompassing bitmap pre-processing, alignment, and slot allocation. Finally, the resulting lightpath configuration is provisioned and presented through the TeraFlow web service component.

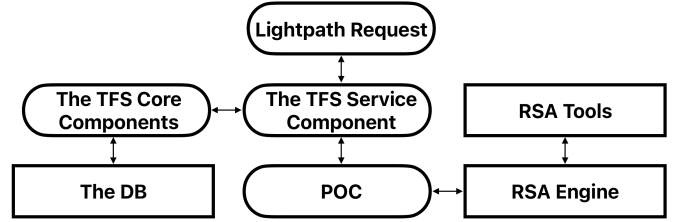


Fig. 3. Service Architecture of the POC within TFS ecosystem.

B. Algorithm

To achieve computationally efficient routing, the algorithm utilizes the NetworkX [13] library for complex network analysis. The RSA Engine converts the initial multi-graph instance G into a simplified graph abstraction, G_s , in which parallel physical links between nodes are collapsed into a single logical edge. By routing over G_s rather than a raw physical multi-graph, the controller prevents the state-space explosion and latency. The algorithm first identifies the shortest path (dsp) via Dijkstra's algorithm and additional candidate path ($dsp + \Delta\mathcal{H}$) satisfying the hop-count constraint $\mathcal{H} \leq \mathcal{H}_{min} + \Delta\mathcal{H}$, where $\mathcal{H}_{min}$ is the shortest path length and $\Delta\mathcal{H}$ is the allowed additional hops. Between these candidate routes, the optimal trajectory is selected and designated as $path_node$. This $path_node$ is then mapped onto the original multi-directed graph G to facilitate parallel path expansion. Following the path selection, the controller initiates the spectrum allocation phase based on FSUs required (N_r) to satisfy the lightpath request. Then it establishes a reference bitmap from the maximum FSU capacity (N_h), found among all endpoint supported FSU capacity (N_s). Each endpoint's N_s is then aligned with N_h to ensure bitwise spectral continuity across links with

heterogeneous frequency ranges. The algorithm is presented as follows:

Algorithm 1 RSA with parallel links

Require: Graph $G = (V, E)$, Links $\mathcal{L}$, Lightpath Request $\mathcal{R}$, Shortest Path Length $\mathcal{H}$ in hops, Allowed Additional Hops $\Delta\mathcal{H}$, Path selection scheme $\mathcal{PS}$, Spectrum selection scheme $\mathcal{SS}$.

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1: Step 1: Create graph  $G$ 
2: for all  $link \in \mathcal{L}$  do
3:   add devices as  $V$ , links as  $E$  in  $G$ 
4: end for
5: Cache the Graph for future re-use
6: Step 2: Path computation
7: convert  $G$  into  $G_s$ 
8:  $dsp \leftarrow find\_path(G_s, \mathcal{PS})$ 
9:  $dsp \leftarrow expand\_parallels(dsp, G, \mathcal{PS})$ 
10:  $(dsp + \Delta\mathcal{H}) \leftarrow find\_path(G_s, \mathcal{H}, \Delta\mathcal{H}, \mathcal{PS})$ 
11:  $(dsp + \Delta\mathcal{H}) \leftarrow expand\_parallels(dsp + \Delta\mathcal{H}, G, \mathcal{PS})$ 
12:  $path\_node \leftarrow dsp \parallel (dsp + \Delta\mathcal{H})$ 
13: Step 3: Perform SA
14:  $B_r \leftarrow calculate\_bandwidth(bitrate, modulation)$ 
15:  $N_r \leftarrow required\_slots(B_r)$ 
16: for all  $hop \in path\_node$  do
17:   get link information
18:   find  $min, max$  frequencies
19: end for
20:  $N_h \leftarrow reference\_bitmap(max, min)$ 
21: for all  $hop \in path\_node$  do
22:   get source, destination endpoint bitmaps
23:   align them with  $N_h$ 
24:   perform intersection & update  $N_h$ 
25: end for
26: apply  $\mathcal{SS}$  on final  $N_h$ 
27:  $\mathcal{R}$  is success or blocked

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IV. PERFORMANCE EVALUATION

To assess the performance of the POC, we performed two distinct evaluations to determine both efficiency and statistical network performance using the **NSFNET** topology. The efficiency evaluation is conducted within the TFS ecosystem by issuing a lightpath request to the controller and recording the execution time in milliseconds across four sequential phases: Graph Generation (GG), Shortest Path Computation (SPC) and Spectrum Assignment (SA). Finally, $\mathcal{E}$ is measured by comparing the execution time using the formula: $\mathcal{E} = NSF_s / NSF_p$, where NSF_s represents the topological configuration with no parallel links followed by NSF_p with parallel links. As

TABLE I
EFFICIENCY EVALUATION OF PARALLELOPTICALCONTROLLER

Topo	GG [ms]	SPC [ms]	SA [ms]	Total [ms]	$\mathcal{E}$
NSF_s	0.34	6.32	4.38	11.05	-
NSF_p	0.44	7.30	4.82	12.57	87.87%

detailed in Table I, execution times naturally increase in

proportion to the structural complexity of the network. Despite this, the system preserves highly competitive efficiency rates peaking at 87.50% across the evaluated configurations.

A. Simulation setup and metrics

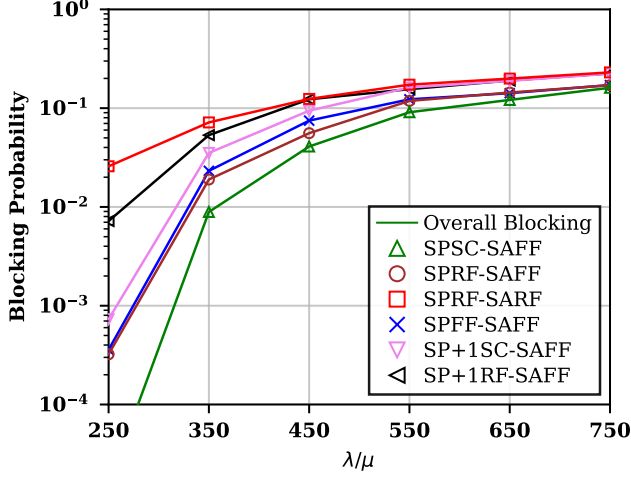
The statistical evaluation was conducted considering NSF_s as the benchmark with 14 Nodes, 22 bi-directional links comparing NSF_p with 14 Nodes, 66 bi-directional links (3x parallel links). Adhering to the ITU-T C-Band flex-grid standard, the spectrum is divided into 350 FSUs featuring specified slot width and central frequency granularity. Experiments were conducted across varying traffic loads using a heterogeneous demand profile: 50% of connection requests at 100 Gbps (requiring 3 FSUs), 25% at 200 Gbps (5 FSUs), and 25% at 400 Gbps (10 FSUs). To perform dynamic traffic simulation, a simulator has been designed using Python's Flask microframework outside the TFS ecosystem. Later, it is containerized using Docker and orchestrated via Docker Compose. Lightpath provisioning requests are dynamically generated following a Poisson process and uniformly distributed among the source-destination pairs. The inter-arrival time and holding time of the lightpath requests are exponentially distributed with an average of $1/\lambda$ and $1/\mu$ seconds, respectively. The load offered to the network in provisioning state condition is expressed in Erlangs as the ratio λ/μ .

B. Simulation results

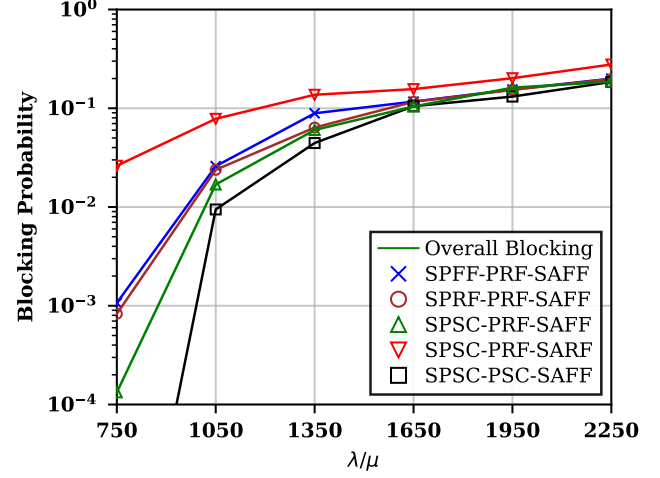
The test cases are designed to evaluate the comparative performance of several path selection schemes within the set of shortest paths (SP): first-fit (SPFF), random-fit (SPRF), and spectrum-check (SPSC). Both RF and SC strategies are further assessed under a configuration allowing for an additional hop beyond the shortest path count (SP+1). Regarding the spectrum assignment (SA), given that first-fit (SAFF) is widely considered the most effective approach, a comparative configuration using random-fit (SARF) is included only as a baseline reference.

The simulation results in Fig. 4(a) demonstrate that the SC scheme consistently achieves superior performance. This outcome is expected, as the scheme evaluates spectral contention using a bottleneck-based strategy. The algorithm traverses each candidate route hop-by-hop to identify the link with the minimum number of available free slots, which dictates the overall bottleneck capacity of that route. After evaluating all candidates, the controller selects the path offering the maximum bottleneck capacity, ensuring traffic is reliably provisioned onto the least congested route. Analysis at 250 Erlangs reveals that SPSC scheme maintains a blocking probability below¹ 10^{-4} , whereas SPFF and SPRF typically range between 10^{-3} and 10^{-4} . Notably, (SP+1SC-SAFF) is outperformed by the shortest-path schemes; this indicates that the inclusion of additional links increases the total resource occupancy, even in the case of spectrum-check. Finally, applying a SARF is confirmed to significantly degrade the blocking probability. The

¹This result is achieved with 72% confidence interval. Even though not very stable, it is obtained with 7M lightpath requests.



(a)



(b)

Fig. 4. P_b vs Erlangs, holding time $1/\mu = 60$ s. Provisioning of the NSFNET topology spectral contention under (a) single-link and (b) $3 \times$ parallel-links.

results in Fig. 4(b) remain consistent in the topology featuring three parallel links under network loads scaled by a factor of three. The SPSC, integrated with parallel spectrum-check (PSC) and spectrum assignment first-fit (SPSC-PSC-SAFF), achieves the most effective results. However, if the number of parallel links increases PSC might introduce exponential execution time because all the candidate paths are recursively explored, thus even if it is not the best performing scheme we are proposing (SPSC-PRF-SAFF) as the optimal solution to be implemented in a real deployment, especially in such scenarios where the computation time is a stringent constraint (e.g., during restoration). The SPFF-PRF and SPRF-PRF exhibit behavior similar to the trends observed in the single-link configuration. Finally, the random-fit spectrum assignment scheme performs poorly even when paired with the superior path selection strategy (SPSC-PRF-SARF), confirming that inefficient spectral allocation significantly increases spectrum fragmentation leading to higher blocking probabilities even if parallel links are available.

V. CONCLUSION & FUTURE WORKS

This work proposed a network controller integrated in the TeraFlow framework and evaluated several RSA schemes utilizing a simplified graph abstraction to manage the computational complexity inherent in parallel-link topologies. The solution was validated through both functional execution time measurements and dynamic simulations. Experimental results demonstrated that the proposed mechanism achieved good computational efficiency and scalability as network density increased. Simulation results demonstrated that SPSC heuristic paired with first-fit spectrum (SPSC-SAFF) allocation emerges as effective and most scalable solution.

Future research will focus on implementing a parallel computation strategy for path selection to reduce the exponential execution time of the spectrum-check strategy to linear time

complexity, thereby establishing the SPSC-PSC scheme as the definitive optimal solution. Additionally, subsequent work will explore advanced frequency modulation techniques and continuous device driver optimization to further enhance the controller's overall performance.

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